Assignment II

Single-Objective and Multi-Objective Clustering

Course: Topics in Data Mining

Total Marks: 20

Submission: Source Code + Report (at most 10 pages) including dataset description, preprocessing, clustering results, evaluation metrics, statistical test results, Pareto front plots, and discussion +PPT

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Part 1: Single-Objective Clustering (10 Marks)

Objective

The aim of this task is to apply and compare two clustering algorithms on datasets from a selected domain and analyze their performance using evaluation metrics and statistical testing.

Do From Scratch:

Step 1: Choose a Domain

Select one application domain where clustering is useful.

Examples include healthcare data, customer segmentation, text document clustering, image clustering, or social media data analysis.

Briefly explain why clustering is useful in the selected domain.

Step 2: Select Two Datasets

Choose two datasets from the same domain. The datasets must be publicly available.

Step 3: Data Preprocessing

Perform possible preprocessing to Prepare the dataset before clustering

- Optional dimensionality reduction (e.g., PCA)

Step 4: Determine the Number of Clusters

Before applying K-Means, determine the optimal number of clusters (k) using at least one method such as:

- Elbow Method

Include the plot used to determine k.

Step 5: Apply K-Means Clustering

Perform K-Means clustering with:

- number of clusters = selected value of k

Step 6: Apply DBSCAN Clustering

Apply DBSCAN clustering where the number of clusters is not required in advance.

Choose parameters epsilon (ε) and MinPts.

Perform 5 runs with different parameter settings.

Step 7: Clustering Evaluation

Evaluate clustering results using internal clustering metric: Silhouette Score

Step 8: Statistical Significance Testing

Apply one statistical test to check whether the clustering algorithms perform significantly differently.

Possible tests include t-test, Wilcoxon Signed-Rank Test, or Friedman Test. Choose one

- Run K-Means 5 times (different random initialization).
- Run DBSCAN 5 times (with different parameter settings)

Step 9: Parameter Sensitivity Analysis

Analyze in report what you observe on parameters changing.

Part 2: Multi-Objective Clustering (10 Marks)

Objective

Formulate clustering as a multi-objective optimization problem.

Do From Scratch:

Step 1: Select Dataset

Use one datasets (same or different from Part 1). And take two objectives and take

All objectives should be: Minimized or Maximized, Normalized in the same scale, Continuous, trade-offs between objectives are linear. Then use Genetic Algorithm.

Step 2: Define Conflicting Objectives

Define two conflicting objectives for clustering.

Example:

1. Cluster compactness (minimize intra-cluster distance)

2. Number of clusters (minimize number of clusters)

Step 3: Multi-Objective Optimization using Genetic Algorithm

Use a Genetic Algorithm for multi-objective clustering.

Possible algorithm: NSGA-II

Step 4: Identify Non-Dominated Solutions

Use Pareto dominance to identify non-dominated clustering solutions.

Step 5: Plot Pareto Front

Plot the Pareto front showing the trade-off between objectives.

Example axes:

X-axis: Number of clusters

Y-axis: Cluster compactness or clustering quality

Note: Plagiarism will result in zero marks. All experiments must be clearly documented.